Analysis of Income distribution in the US (1994)

1. Project Overview and Objective

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to derive meaningful insights.

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2. Data Sources

Source Description and Timeline: Data source link **and** 1994.

- **Domain**: Census

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3. Problem Statement

- To analyse income status across a select population based on census data to identify and categorize their income into two i.e., above 50K and less than or equal to 50K.
- To study various factors and visualize how each factor affect the income of this select population.
- To determine the most and the least contributing factors.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Age	Numeric	Age of the individual during the census
Work class	Categorical	Sector in which the individual works
Education	Categorical	Highest education of the individual
Education number	Numerical	Total no. of years spent for education
Marital Status	Categorical	Mentions the marital status of the individual
Occupation	Categorical	Mentions the occupation of the individual
Relationship	Categorical	Mentions the relationship of the individual
Ethnicity	Categorical	Mentions the ethnicity of the individual

Gender	Categorical	Mentions the gender of the individual
Hours per week	Numerical	Number of hours the individual works
Native country	Categorical	Native country of the individual
Income	Numerical	Mentions whether the individual's income is above 50K or equal to or below 50K dollars

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, and standardized formats.
- Created a calculated column based on age using the formula: *=IF([@Age]>50,"Senior",IF([@Age]>30,"Middle aged","Youth"))*
- Created a calculated column based on working hours using the formula: *=IFS([@[Hours per week]]<=30,"Low working hours",[@[Hours per week]]<=60,"Average working hours",[@[Hours per week]]<=99,"High working hours")*
- Created a calculated column based on income using the formula: *=IF([@Income]="<=50K","Low income","High income")*
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.

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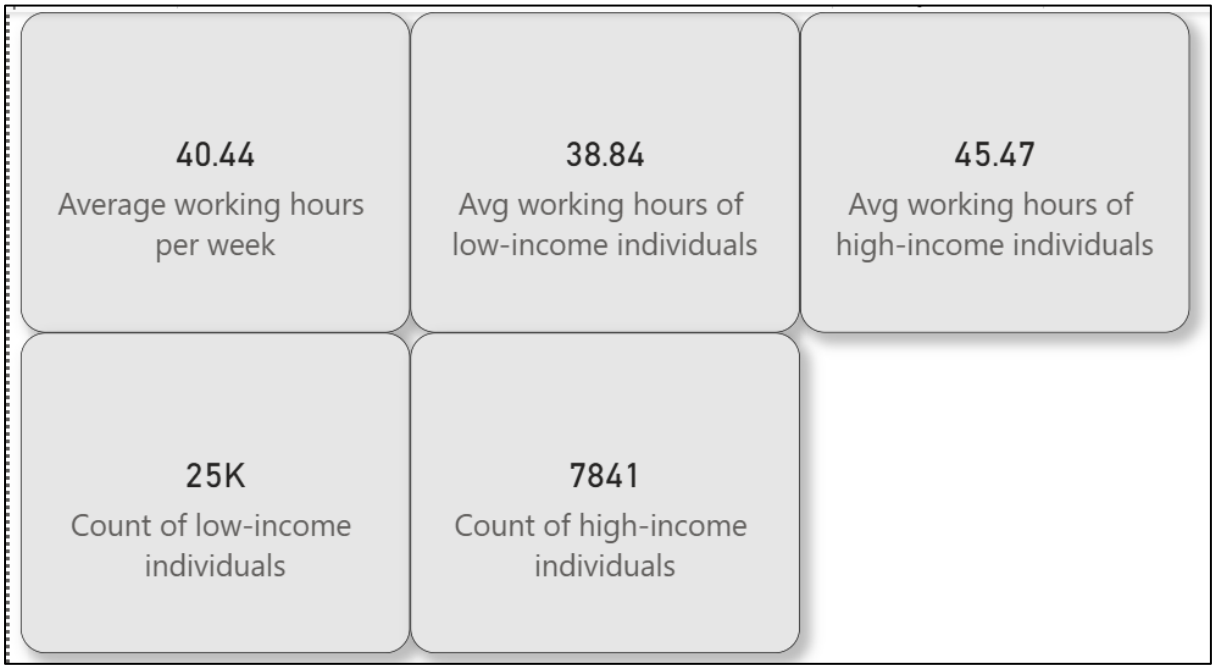
7. Data Modelling and DAX (Power BI)

- **Calculated DAX Measures:** Implemented DAX formulas for key metrics, such as average working hours and number high- and low-income individuals.



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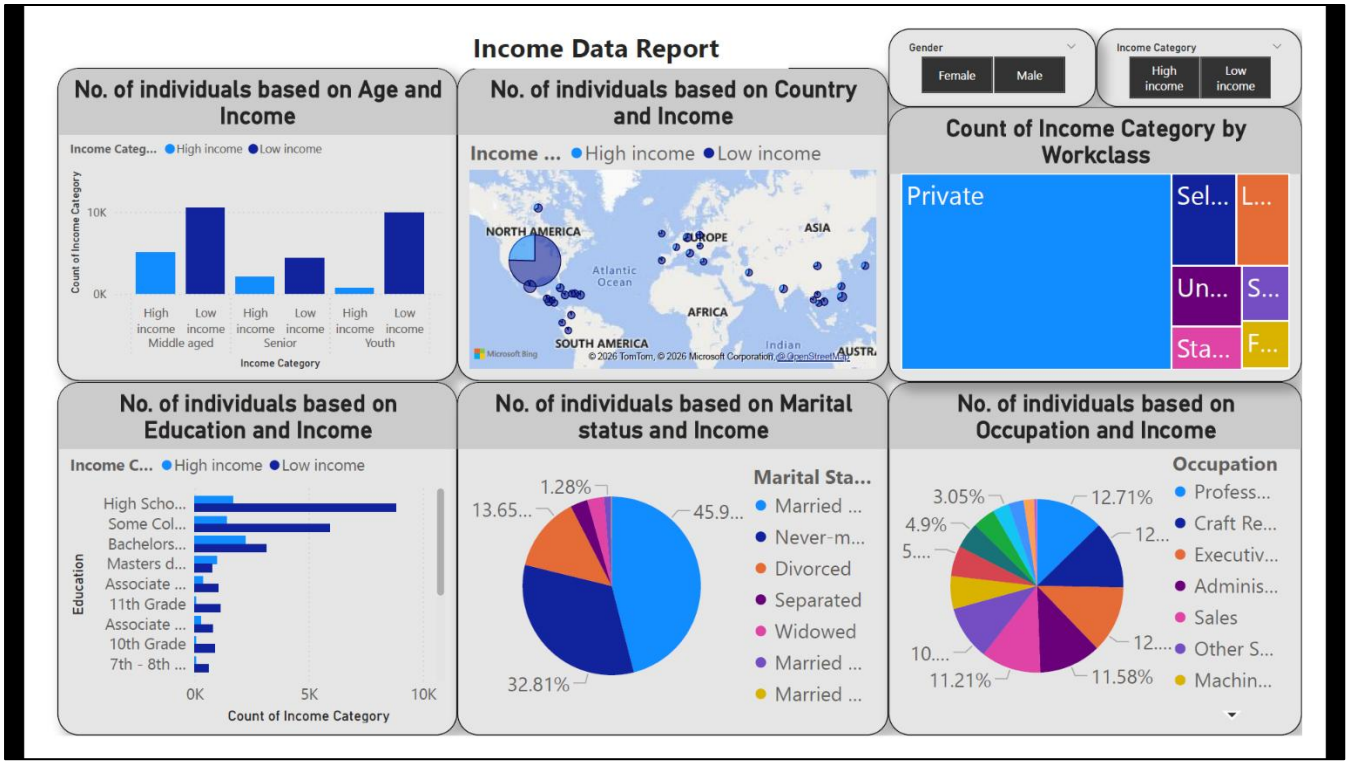
8. Analysis and Visualizations (Power BI)

Dashboard:



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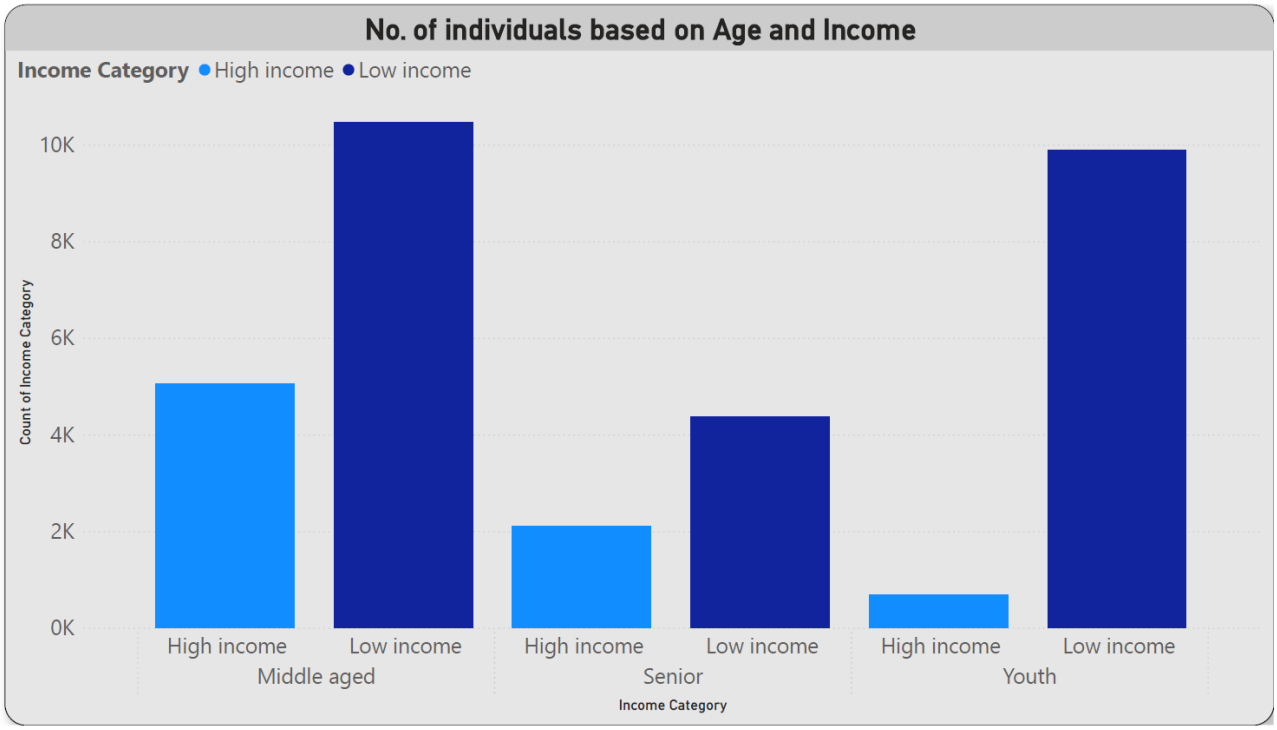
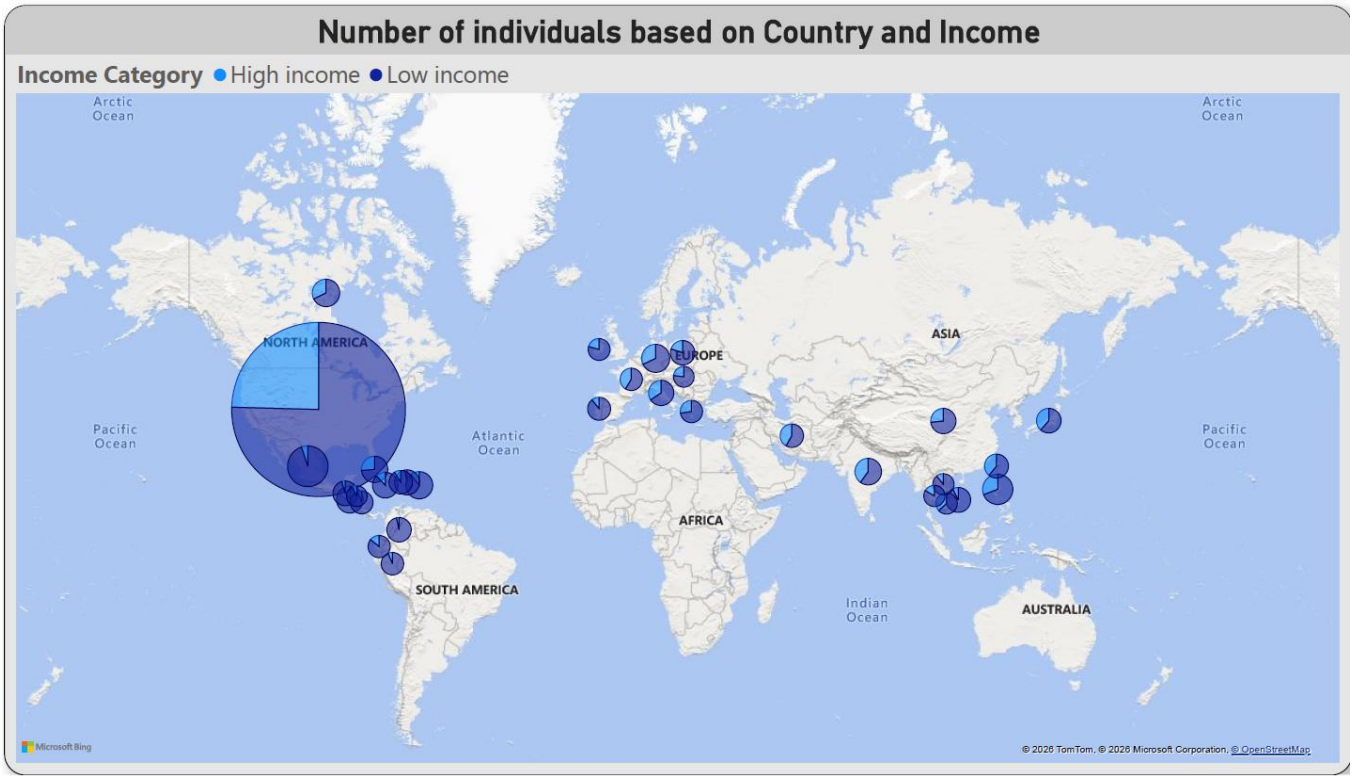
9. Insights & Conclusions

- According to the data 75.92% of the people have low income and 24.08% of the people have high income.

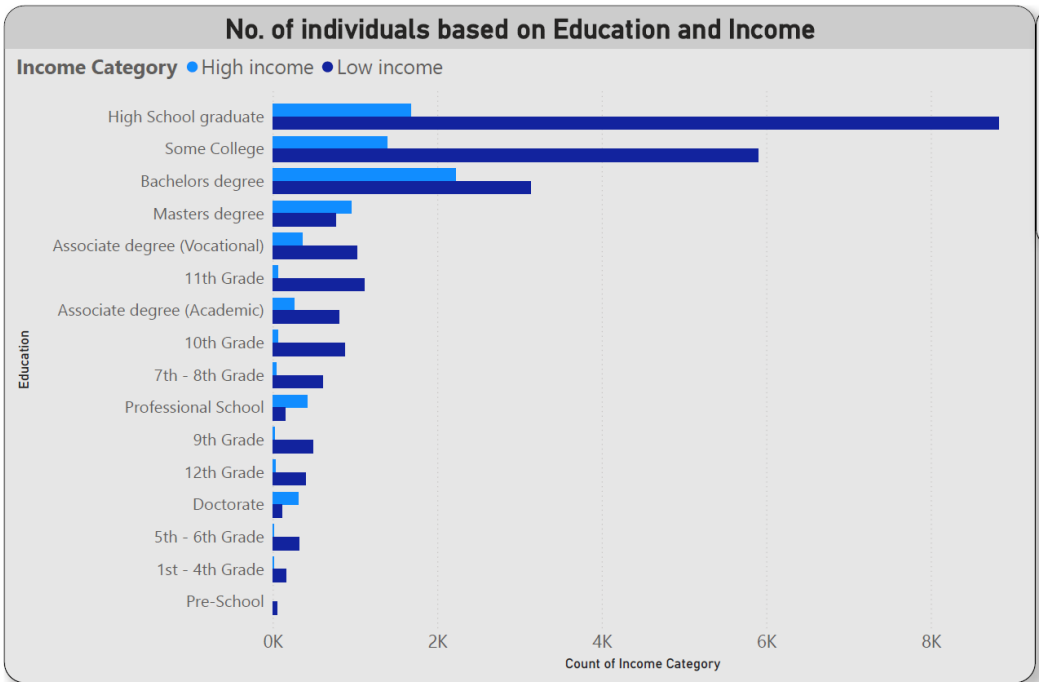
- The chart below shows that the high-income individuals are low in number compared to low-income individuals across all age groups. The difference is very pronounced particularly among the youth.



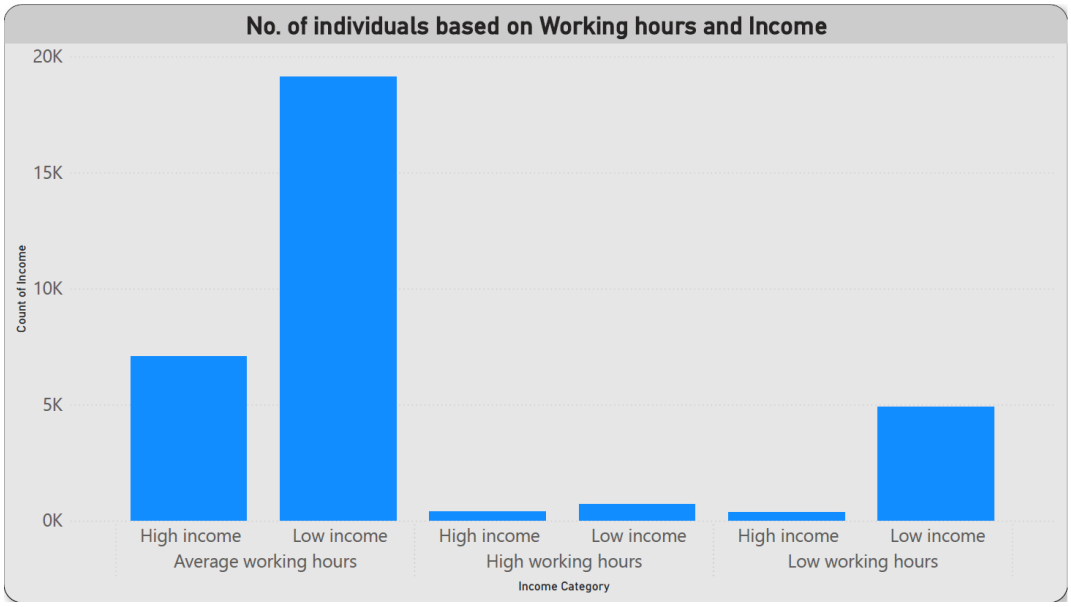
- As per the chart below, this dataset includes people with their origin in various countries across the globe. However, the majority of them has the US as their native country. Almost all other countries contribute more or less same number of individuals and their income distribution is also similar.



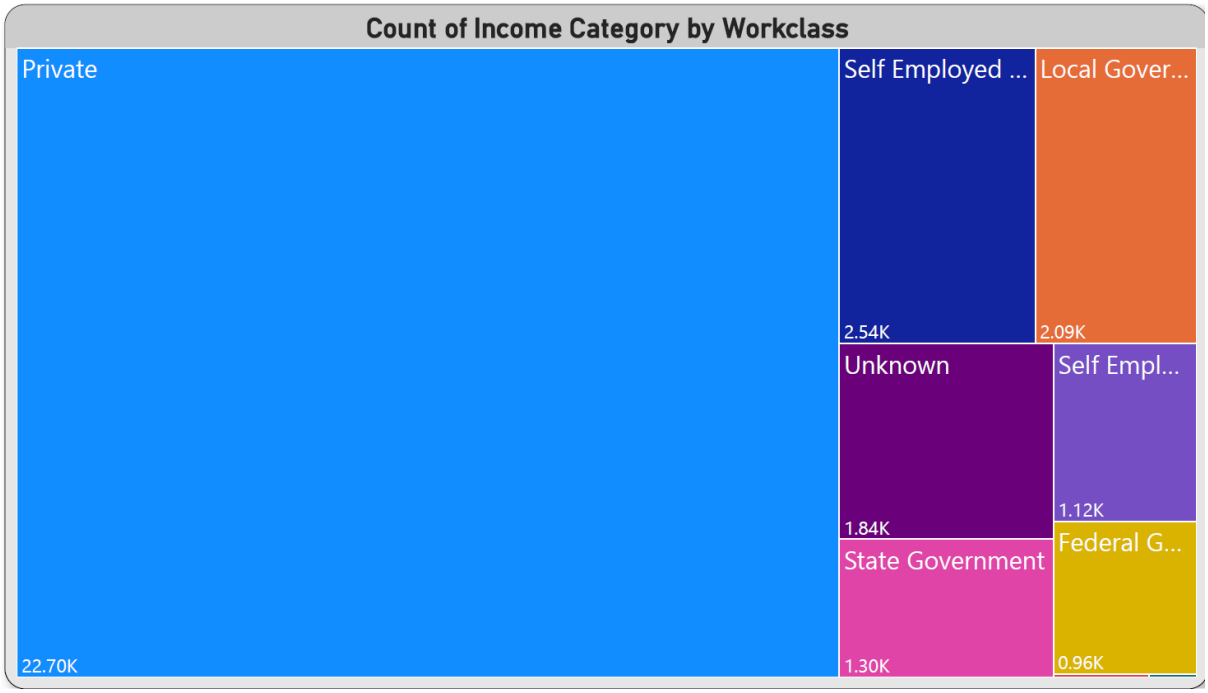
- While analysing the income across various education levels, the below chart shows that the higher the education, higher is the individual's income. Most of the people in this dataset are high school graduates followed by individuals those who completed some college.
- Most of the categories have the no. of low-income individuals more than the high-income individuals except three categories. They are Doctorate holders, Master degree holders and people who completed professional school. These three categories have high no. of high-income individuals than the low-income individuals.



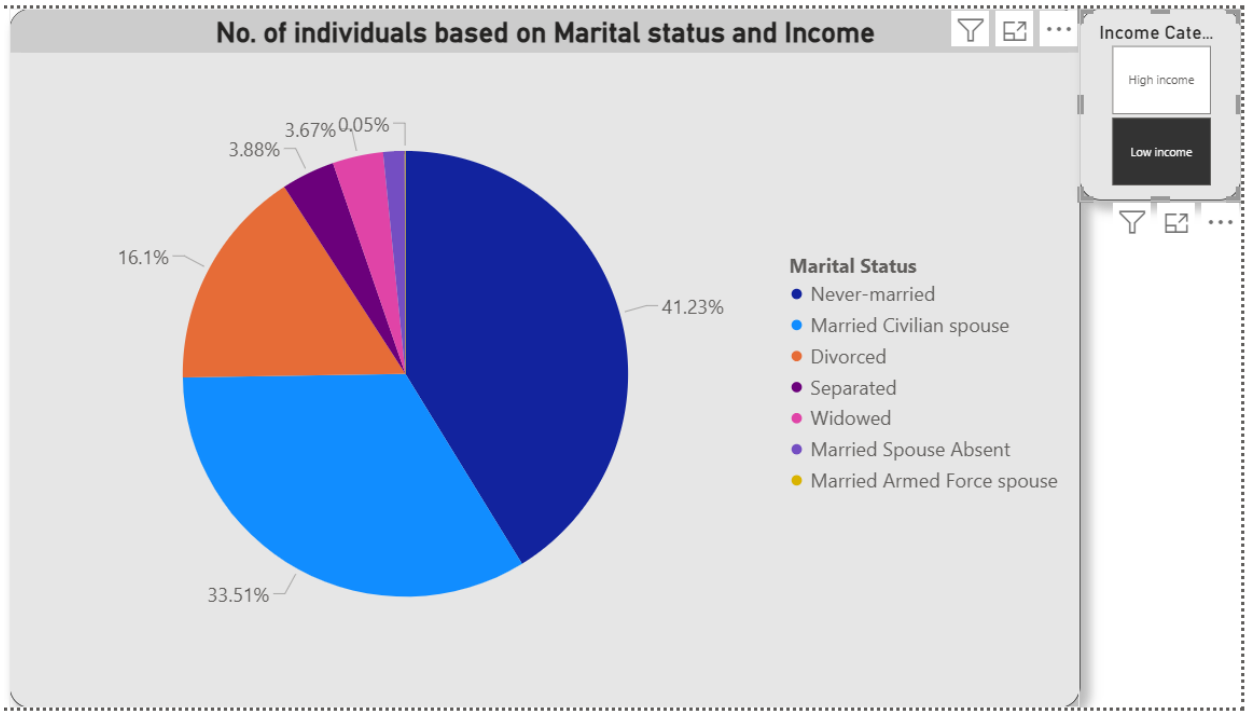
- A large number of people have average working hours (30-40 hours/week). The income gap is significant among the people with average and low working hours. However, the gap is marginal among the people with high working hours.



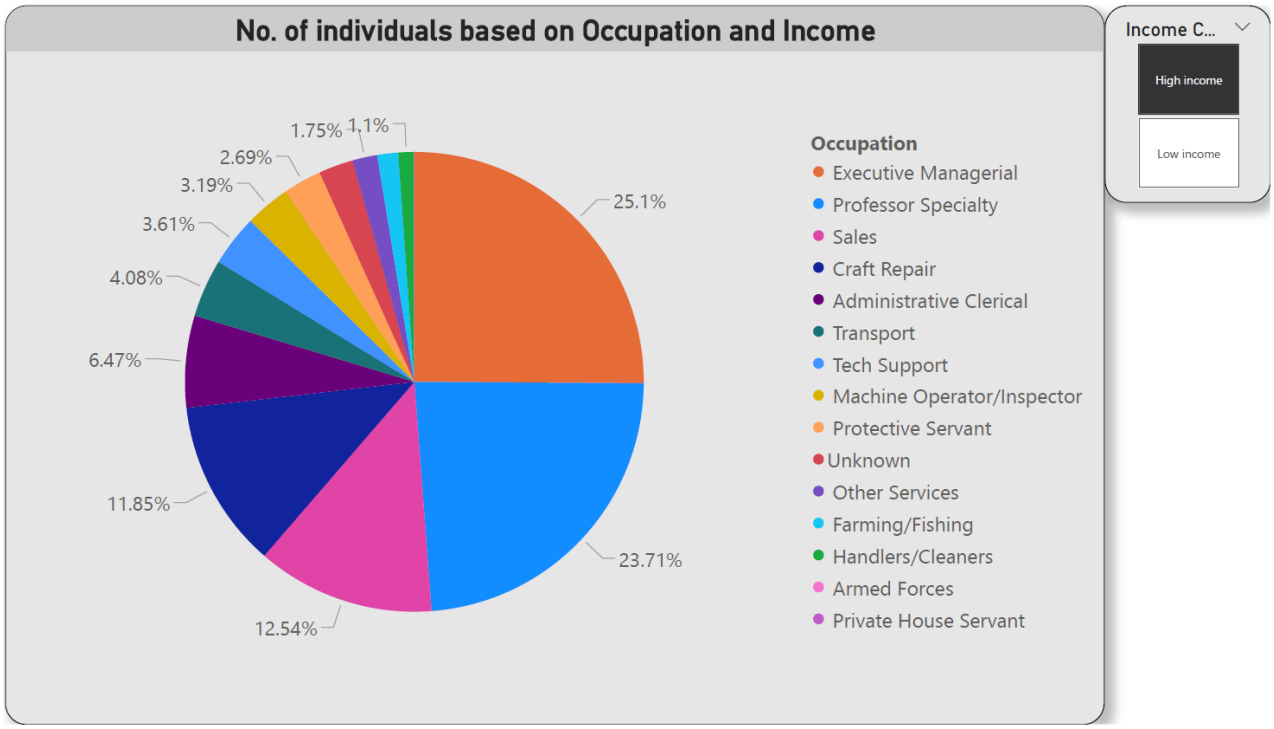
- The tree map below shows the distribution of the people across various work class. Around 70% of the people work in Private sector, followed by people who are self employed and people working in local Government. All other categories are evenly spread.



- While analysing the low-income distribution across marital status, it is clear that 41.23% of the low-income individuals are unmarried. This is followed by the individuals who married civilian spouse and divorced individuals.



- The pie chart below shows the distribution of high-income individuals based on occupation. Executive managers rank first, followed by Professors, Sales professionals and Craft repair people. The rest of the occupations are evenly spread with their individual shares less than 10%.



10. Conclusion

The charts were useful in analysing the income status across the select population based on census data to identify and categorize income.

Studied various factors and visualized how each factor affected the income of the population.

The most contributing factors are Marital status, Occupation, Working hours and Education. Married people have higher income than unmarried, separated and divorced individuals. Education and Occupation also play a major role.

The least contributing factors are Native country and Age. There is no considerable difference based on these factors.